

 Azure Security Incident Report

Generated by Azure Adversary Simulation Lab | Workspace: security-workspace

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# SECURITY INCIDENT REPORT
**Incident ID:** INC-2026-0522-001
**Date Generated:** 2026-05-23
**Classification:** CONFIDENTIAL
**Analyst:** Senior SOC Analyst
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1. EXECUTIVE SUMMARY

A coordinated credential-based attack originating from IP `24.243.129.70` targeted the Azure tenant `fernandocop12gmail.onmicrosoft.com`, progressing from failed authentication attempts against a user account to successful enumeration and exfiltration of secrets from Azure Key Vault `vulnkeyvault2026`. The attacker successfully retrieved the `DatabasePassword` secret within approximately 13 minutes of initial failed sign-in attempts, indicating either credential compromise through an alternate vector or token theft. This represents a confirmed data breach of privileged credentials with potential for lateral movement to database systems.

2. ATTACK TIMELINE

Timestamp (UTC)	Event	Status	Source IP	Interpretation
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2026-05-22 23:34:42.445	`VaultGet` - Key Vault metadata enumeration	Success	20.119.29.106	Reconnaissance phase - Attacker enumerated vault configuration via ARM API (different IP suggests Azure Cloud Shell or separate infrastructure) 
2026-05-22 23:40:34.771	Failed sign-in (victim-user) - Error 50126	Failed	24.243.129.70	Initial access attempt - Password spray/brute force initiated 
2026-05-22 23:40:55.490	Failed sign-in (victim-user) - Error 50126	Failed	24.243.129.70	Continued brute force - 21-second interval between attempts 
2026-05-22 23:54:03.732	Key Vault SecretList authentication	Failed		

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Unauthorized | 24.243.129.70 | **Pivot attempt** - Attacker testing alternate
access vector |
| **2026-05-22 23:54:04.609** | `SecretList` - Enumerate all secrets | ✓
Success | 24.243.129.70 | **Collection phase** - Successfully listed vault
contents |
| **2026-05-22 23:54:05.490** | Key Vault SecretGet authentication | ✗
Unauthorized | 24.243.129.70 | Transient auth failure (token refresh) |
| **2026-05-22 23:54:05.745** | `SecretGet` - DatabasePassword retrieved | ✓
Success | 24.243.129.70 | **EXFILTRATION CONFIRMED** - Sensitive credential
extracted |
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**Total Attack Duration:** ~19 minutes 23 seconds
**Time from recon to exfiltration:** ~19 minutes
**Gap between failed sign-ins and successful KV access:** ~13 minutes
(credential compromise occurred in this window)
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3. MITRE ATT&CK TTPs IDENTIFIED

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| Tactic | Technique ID | Technique Name | Evidence |
|-----|-----|-----|-----|
| **Reconnaissance** | T1526 | Cloud Service Discovery | `VaultGet` operation
enumerating vault metadata via ARM API |
| **Initial Access** | T1110.003 | Brute Force: Password Spraying | Two failed
sign-ins with 21-sec interval, same IP, Error 50126 |
| **Initial Access** | T1078.004 | Valid Accounts: Cloud Accounts | Successful
Key Vault access implies compromised credentials |
| **Credential Access** | T1552.001 | Unsecured Credentials: Credentials in
Files | Targeting secrets stored in Key Vault |
| **Discovery** | T1087.004 | Account Discovery: Cloud Account | Enumeration
targeting specific user account |
| **Collection** | T1213.003 | Data from Cloud Storage | `SecretList` +
`SecretGet` operations on Key Vault |
| **Exfiltration** | T1041 | Exfiltration Over C2 Channel | Database credentials
extracted to attacker-controlled IP |
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4. SEVERITY ASSESSMENT

SEVERITY: CRITICAL ●

Justification:

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| Factor | Assessment | Impact |
|-----|-----|-----|
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| **Data Classification** | Database credentials exfiltrated | Potential access
to production data stores |
| **Blast Radius** | Key Vault secrets typically protect tier-0 assets |
Cascading compromise risk |
| **Confirmed Exfiltration** | `SecretGet` for `DatabasePassword` returned "OK"
| Active breach, not attempt |
| **Attacker Persistence** | Unknown; access may be retained | Ongoing risk |
| **Regulatory Impact** | Database access may contain PII/PHI | Potential
GDPR/HIPAA notification required |
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****Risk Score:**** 9.2/10 (CVSS-style assessment)

5. DETECTION GAPS ANALYSIS

❌ What Was NOT Detected/Prevented:

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| Gap | Why It Matters | Root Cause |
|-----|-----|-----|
| **No account lockout triggered** | Only 2 failed attempts logged; attacker may
have succeeded elsewhere or used token theft | Smart lockout threshold likely
set >10 attempts; password spray stayed under radar |
| **Missing successful sign-in log** | No record of how attacker obtained valid
session for Key Vault | Possible token theft (T1528), pass-the-token attack, or
compromised service principal |
| **VaultGet from different IP not correlated** | `20.119.29.106` (Azure
datacenter IP) performed recon 6 mins before attack | No cross-IP behavioral
correlation; Azure-internal IPs may bypass scrutiny |
| **No impossible travel alert** | No alert despite different IPs accessing
resources | May indicate Cloud Shell abuse or IP reputation not flagged |
| **SecretGet not blocked despite prior Unauthorized** | Attacker succeeded
after failed auth attempts | No adaptive access control blocking suspicious
patterns |
| **No Key Vault Firewall alerts** | External IP successfully accessed Key Vault
| Key Vault likely has no network restrictions configured |
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🔍 Critical Missing Telemetry:

- Azure AD successful sign-in logs (filtered out or not ingested?)
- Service Principal authentication logs
- Conditional Access policy evaluation logs
- Azure Activity Logs showing RBAC assignments

6. IMMEDIATE RESPONSE ACTIONS

🚨 PRIORITY 1 - NOW (0-30 minutes)

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- 1.1 ROTATE the compromised DatabasePassword secret IMMEDIATELY
az keyvault secret set --vault-name vulnkeyvault2026 --name DatabasePassword --value ""
- 1.2 DISABLE/RESET victim-user account
Set-AzureADUser -ObjectId "victim-user@fernandocop12gmail.onmicrosoft.com" -AccountEnabled \$false
Revoke-AzureADUserAllRefreshToken -ObjectId ""
- 1.3 BLOCK attacker IP at Azure Firewall/NSG
 - Block 24.243.129.70 (primary attack source)
 - Investigate 20.119.29.106 (recon source - Azure IP, may be legitimate)
- 1.4 Enable Key Vault firewall - deny all public access
az keyvault update --name vulnkeyvault2026 --default-action Deny

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💡 PRIORITY 2 - URGENT (30 mins - 2 hours)

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- 2.1 Review ALL secrets in vulnkeyvault2026 - assume compromised if accessed
az keyvault secret list --vault-name vulnkeyvault2026
- 2.2 Audit Key Vault access policies - identify over-permissioned identities
az keyvault show --name vulnkeyvault2026 --query "properties.accessPolicies"
- 2.3 Search for lateral movement - query database audit logs for:
 - Connections using compromised credentials
 - Unusual query patterns